

GrounUp Enterprise Specification

Phase 14 - HSE, Safety, Environmental, Quality & Compliance

Version 1.0.0 | Review-ready | Proposed lock: GES-P14-v1.0.0

Mission

Create an enterprise HSEQ system that protects workers and the public, manages environmental and quality obligations, preserves evidence and enforces critical controls.

Design Principles

Mobile-first; stop-work authority; competency controls; evidence-first compliance; risk escalation; auditability; configurable jurisdiction rules; human authority for safety-critical decisions.

Functional Domains

Safety Management

Safety plans, observations, hazards, controls, inspections and field safety execution.

Incident & Near-Miss Management

Incident intake, classification, investigation, root cause, corrective actions and regulatory readiness.

JHA/JSA & Pre-Task Planning

Job hazard analysis, daily pre-task plans, crew acknowledgement and changing-condition controls.

Permits to Work

Confined space, hot work, excavation, lifting, electrical and other controlled-work permits.

Training & Competency

Required training, toolbox talks, qualifications, expirations and competency verification.

Environmental Management

Environmental plans, inspections, spills, stormwater, erosion control, waste and permit obligations.

Quality Management

Inspection/test plans, hold points, NCRs, punch items, corrective action and quality records.

Compliance & Regulatory

Regulatory obligations, inspections, evidence, deadlines, citations and compliance calendars.

Industrial Hygiene

Exposure assessments, monitoring, PPE programs, respiratory/hearing controls and occupational hygiene.

Emergency Management

Emergency action plans, drills, contacts, response roles and post-event review.

Subcontractor HSEQ

Prequalification, orientation, compliance, observations, incidents and performance scorecards.

Equipment & Fleet Safety

Daily inspections, lockout/tagout interfaces, operator authorization and equipment safety controls.

Public & Traffic Safety

MOT/traffic control, public protection, pedestrian interfaces and site-access safety.

Corrective & Preventive Action

CAPA ownership, due dates, verification, recurrence monitoring and effectiveness review.

Audit & Assurance

Internal audits, field audits, management reviews, findings, evidence and closure.

HSEQ Analytics

Leading/lagging indicators, trends, risk heat maps, recurrence, quality and environmental performance.

AI HSEQ Intelligence

Evidence-grounded assistance for hazard recognition, classification, trend analysis and recommendations.

HSEQ Digital Twin

Current safety, environmental, quality and compliance state linked to projects, crews, assets and locations.

Controlled Artifact Summary

Artifact	Count
Requirements	320
Entities	145
Fields	219
Business Rules	60
Validation Rules	60
Workflows	35
Engines	40
Api Resources	50
Ui Surfaces	35
Reports	30
Kpis	45
Tests	320

KPI Registry

P14-KPI-001 - TRIR - Safety Management - Governed formula/definition for TRIR.

P14-KPI-002 - DART Rate - Incident & Near-Miss Management - Governed formula/definition for DART Rate.

P14-KPI-003 - Lost Time Incident Rate - JHA/JSA & Pre-Task Planning - Governed formula/definition for Lost Time Incident Rate.

P14-KPI-004 - Near Miss Reporting Rate - Permits to Work - Governed formula/definition for Near Miss Reporting Rate.

P14-KPI-005 - Safety Observation Rate - Training & Competency - Governed formula/definition for Safety Observation Rate.

P14-KPI-006 - Positive Observation % - Environmental Management - Governed formula/definition for Positive Observation %.

P14-KPI-007 - Hazard Closure Time - Quality Management - Governed formula/definition for Hazard Closure Time.

P14-KPI-008 - Critical Hazard Open Count - Compliance & Regulatory - Governed formula/definition for Critical Hazard Open Count.

P14-KPI-009 - Stop Work Events - Industrial Hygiene - Governed formula/definition for Stop Work Events.

P14-KPI-010 - JHA Completion % - Emergency Management - Governed formula/definition for JHA Completion %.

P14-KPI-011 - Pre-Task Plan Compliance - Subcontractor HSEQ - Governed formula/definition for Pre-Task Plan Compliance.

P14-KPI-012 - Toolbox Talk Completion - Equipment & Fleet Safety - Governed formula/definition for Toolbox Talk Completion.

P14-KPI-013 - Training Compliance % - Public & Traffic Safety - Governed formula/definition for Training Compliance %.

P14-KPI-014 - Certification Compliance % - Corrective & Preventive Action - Governed formula/definition for Certification Compliance %.

P14-KPI-015 - Inspection Completion % - Audit & Assurance - Governed formula/definition for Inspection Completion %.

P14-KPI-016 - Inspection Finding Closure - HSEQ Analytics - Governed formula/definition for Inspection Finding Closure.

P14-KPI-017 - Corrective Action On-Time % - AI HSEQ Intelligence - Governed formula/definition for Corrective Action On-Time %.

P14-KPI-018 - Corrective Action Effectiveness % - HSEQ Digital Twin - Governed formula/definition for Corrective Action Effectiveness %.

P14-KPI-019 - Incident Investigation Cycle Time - Safety Management - Governed formula/definition for Incident Investigation Cycle Time.

P14-KPI-020 - Repeat Incident Rate - Incident & Near-Miss Management - Governed formula/definition for Repeat Incident Rate.

P14-KPI-021 - Environmental Inspection Compliance - JHA/JSA & Pre-Task Planning - Governed formula/definition for Environmental Inspection Compliance.

P14-KPI-022 - SWPPP Compliance - Permits to Work - Governed formula/definition for SWPPP Compliance.

P14-KPI-023 - Spill Count - Training & Competency - Governed formula/definition for Spill Count.

P14-KPI-024 - Spill Volume - Environmental Management - Governed formula/definition for Spill Volume.

P14-KPI-025 - Waste Manifest Compliance - Quality Management - Governed formula/definition for Waste Manifest Compliance.

P14-KPI-026 - Quality First-Pass Yield - Compliance & Regulatory - Governed formula/definition for Quality First-Pass Yield.

P14-KPI-027 - NCR Rate - Industrial Hygiene - Governed formula/definition for NCR Rate.

P14-KPI-028 - Rework Cost % - Emergency Management - Governed formula/definition for Rework Cost %.

P14-KPI-029 - Hold Point Compliance - Subcontractor HSEQ - Governed formula/definition for Hold Point Compliance.

P14-KPI-030 - Test Pass Rate - Equipment & Fleet Safety - Governed formula/definition for Test Pass Rate.

P14-KPI-031 - Punch Closure Time - Public & Traffic Safety - Governed formula/definition for Punch Closure Time.

P14-KPI-032 - Audit Finding Closure - Corrective & Preventive Action - Governed formula/definition for Audit Finding Closure.

P14-KPI-033 - Regulatory Obligation Compliance - Audit & Assurance - Governed formula/definition for Regulatory Obligation Compliance.

P14-KPI-034 - Subcontractor HSEQ Score - HSEQ Analytics - Governed formula/definition for Subcontractor HSEQ Score.

P14-KPI-035 - Fleet Safety Inspection Compliance - AI HSEQ Intelligence - Governed formula/definition for Fleet Safety Inspection Compliance.

P14-KPI-036 - Operator Authorization Compliance - HSEQ Digital Twin - Governed formula/definition for Operator Authorization Compliance.

P14-KPI-037 - Traffic Control Inspection Compliance - Safety Management - Governed formula/definition for Traffic Control Inspection Compliance.

P14-KPI-038 - Emergency Drill Completion - Incident & Near-Miss Management - Governed formula/definition for Emergency Drill Completion.

P14-KPI-039 - Industrial Hygiene Compliance - JHA/JSA & Pre-Task Planning - Governed formula/definition for Industrial Hygiene Compliance.

P14-KPI-040 - HSEQ Leading Indicator Index - Permits to Work - Governed formula/definition for HSEQ Leading Indicator Index.

P14-KPI-041 - HSEQ Risk Index - Training & Competency - Governed formula/definition for HSEQ Risk Index.

P14-KPI-042 - HSEQ Cost Rate - Environmental Management - Governed formula/definition for HSEQ Cost Rate.

P14-KPI-043 - HSEQ AI Recommendation Acceptance - Quality Management - Governed formula/definition for HSEQ AI Recommendation Acceptance.

P14-KPI-044 - HSEQ AI Recommendation Benefit - Compliance & Regulatory - Governed formula/definition for HSEQ AI Recommendation Benefit.

P14-KPI-045 - HSEQ Digital Twin Freshness - Industrial Hygiene - Governed formula/definition for HSEQ Digital Twin Freshness.

Engine Registry

P14-ENG-001 - Risk Scoring Engine - Deterministic - Provides risk scoring engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-002 - Hazard Control Engine - Deterministic - Provides hazard control engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-003 - JHA Completeness Engine - Deterministic - Provides jha completeness engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-004 - Pre-Task Plan Engine - Deterministic - Provides pre-task plan engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-005 - Permit Eligibility Engine - Deterministic - Provides permit eligibility engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-006 - Stop Work Control Engine - Deterministic - Provides stop work control engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-007 - Incident Classification Engine - Deterministic - Provides incident classification engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-008 - Recordability Decision Support - Deterministic - Provides recordability decision support with governed inputs, auditable outputs and explicit human authority.

P14-ENG-009 - Root Cause Assistant - Deterministic - Provides root cause assistant with governed inputs, auditable outputs and explicit human authority.

P14-ENG-010 - CAPA Engine - Deterministic - Provides capa engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-011 - Corrective Action Escalation Engine - Deterministic - Provides corrective action escalation engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-012 - Training Compliance Engine - Deterministic - Provides training compliance engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-013 - Certification Expiry Engine - Deterministic - Provides certification expiry engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-014 - Competency Verification Engine - Deterministic - Provides competency verification engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-015 - Inspection Scheduling Engine - Deterministic - Provides inspection scheduling engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-016 - Inspection Finding Engine - Deterministic - Provides inspection finding engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-017 - Environmental Compliance Engine - Deterministic - Provides environmental compliance engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-018 - SWPPP Compliance Engine - Deterministic - Provides swppp compliance engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-019 - Spill Response Engine - Deterministic - Provides spill response engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-020 - Waste Compliance Engine - Deterministic - Provides waste compliance engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-021 - Industrial Hygiene Engine - Deterministic - Provides industrial hygiene engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-022 - Quality ITP Engine - Deterministic - Provides quality itp engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-023 - NCR Engine - Deterministic - Provides ncr engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-024 - Hold Point Engine - Deterministic - Provides hold point engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-025 - Test Result Validation Engine - Deterministic - Provides test result validation engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-026 - Subcontractor HSEQ Risk Engine - Deterministic - Provides subcontractor hseq risk engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-027 - Fleet Safety Engine - Deterministic - Provides fleet safety engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-028 - Operator Authorization Engine - Deterministic - Provides operator authorization engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-029 - Traffic/Public Safety Engine - Deterministic - Provides traffic/public safety engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-030 - Regulatory Calendar Engine - Deterministic - Provides regulatory calendar engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-031 - Audit Assurance Engine - Deterministic - Provides audit assurance engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-032 - HSEQ KPI Engine - Deterministic - Provides hseq kpi engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-033 - Trend & Recurrence Engine - Deterministic - Provides trend & recurrence engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-034 - HSEQ Anomaly Detection Engine - AI-assisted - Provides hseq anomaly detection engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-035 - HSEQ Executive Narrative Engine - AI-assisted - Provides hseq executive narrative engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-036 - HSEQ Recommendation Engine - AI-assisted - Provides hseq recommendation engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-037 - Evidence Retrieval Engine - AI-assisted - Provides evidence retrieval engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-038 - HSEQ Confidence Engine - AI-assisted - Provides hseq confidence engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-039 - HSEQ Digital Twin Engine - AI-assisted - Provides hseq digital twin engine with governed inputs, auditable outputs and explicit human authority.

P14-ENG-040 - Emergency Readiness Engine - AI-assisted - Provides emergency readiness engine with governed inputs, auditable outputs and explicit human authority.

Workflow Registry

P14-WF-001 - Hazard Identification & Control - Safety Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-002 - JHA/JSA Creation & Approval - Incident & Near-Miss Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-003 - Daily Pre-Task Planning - JHA/JSA & Pre-Task Planning - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-004 - Toolbox Talk & Acknowledgement - Permits to Work - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-005 - Safety Observation - Training & Competency - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-006 - Stop Work & Release - Environmental Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-007 - Incident Intake - Quality Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-008 - Incident Triage - Compliance & Regulatory - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-009 - Incident Investigation - Industrial Hygiene - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-010 - Root Cause & CAPA - Emergency Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-011 - Regulatory Notification - Subcontractor HSEQ - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-012 - Safety Inspection - Equipment & Fleet Safety - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-013 - Environmental Inspection - Public & Traffic Safety - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-014 - SWPPP Inspection - Corrective & Preventive Action - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-015 - Spill Response - Audit & Assurance - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-016 - Waste Manifest - HSEQ Analytics - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-017 - Permit-to-Work - AI HSEQ Intelligence - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-018 - Confined Space Entry - HSEQ Digital Twin - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-019 - Hot Work - Safety Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-020 - Excavation Permit - Incident & Near-Miss Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-021 - Lift Plan & Permit - JHA/JSA & Pre-Task Planning - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-022 - LOTO / Energy Isolation - Permits to Work - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-023 - Training Assignment & Completion - Training & Competency - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-024 - Certification Expiration - Environmental Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-025 - Subcontractor HSEQ Prequalification - Quality Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-026 - Quality Inspection & Test - Compliance & Regulatory - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-027 - NCR & Corrective Action - Industrial Hygiene - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-028 - Hold/Witness Point Release - Emergency Management - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-029 - Punch/Deficiency Closure - Subcontractor HSEQ - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-030 - Internal Audit - Equipment & Fleet Safety - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-031 - Regulatory Audit - Public & Traffic Safety - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-032 - Emergency Drill/Event - Corrective & Preventive Action - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-033 - Industrial Hygiene Monitoring - Audit & Assurance - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-034 - Management Review - HSEQ Analytics - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

P14-WF-035 - HSEQ Digital Twin Refresh - AI HSEQ Intelligence - Initiate > Assess > Control/Investigate > Review > Approve > Verify > Close

Safety-Critical Governance

AI cannot override stop-work, competent-person, regulated notification, quality hold-point or other configured safety-critical controls. Consequential HSEQ recommendations require evidence, confidence and authorized human review.